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INVITED ARTICLE

ALife approach to eco-evo-devo using evolution of virtual creatures

Takaya Arita[1] · Michał Joachimczak[1] · Takashi Ito[1] · Atsushi Asakura[1] · Reiji Suzuki[1]

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Abstract Evolutionary developmental biology, *evo-devo*, emerged to integrate evolution and development in the 1980s, in which evolution is conceptualized as heritable changes in development. Recently, the field is moving to a new synthesis: ecological evolutionary developmental biology, *eco-evo-devo*. We believe that artificial life (ALife) approach will provide new insights into the field, and also contribute to the field of robotics through the emergence perspective and the constructive methodology. This paper explores the potential of such an artificial life approach based on the evolution of virtual creatures by presenting our ongoing studies with three models: (1) metamorphosis model, (2) exaptation model and (3) Prey-predator model.

Keywords Artificial life · Eco-evolutionary feedback · Eco-evo-devo · Exaptation · Metamorphosis · Morphid academy

1 Introduction

The modern synthesis, developed in the 1930s and 1940s, refers to the theoretical fusion of Darwinian natural selection and Mendelian genetics. It is considered one of the greatest achievements of twentieth-century science. At the same time, it is increasingly recognized that the modern synthesis is limited in its neglect of ontogenetic development. In the early 1980s, evolutionary developmental biology [1, 2], *evo-devo*, emerged to address the neglect of development in the standard framework of evolutionary theory, with the methodological advances such as techniques for gene cloning and visualization of gene activity. Evo-devo views evolution as the results of inherited changes in the pattern of development. Recently, the field is moving to a new synthesis: ecological evolutionary developmental biology, *eco-evo-devo* [3]. This field acknowledges the fact that organisms are continually subject to a changing environment, whether it be changes in nutrition, temperature, predators, competitors or all of these simultaneously [4].

What are the key questions of ecological evolutionary developmental biology? Müller [5] poses a series of core questions classified into (a) evo-devo, (b) devo-evo and (c) eco-evo-devo questions as follows. (a1) How did development originate?, (a2) How did the developmental repertoire evolve?, and (a3) How are developmental processes modified in evolution? (b1) How does development influence phenotypic variation?, (b2) How does development contribute to phenotypic novelty?, and (b3) How does development affect the organization of phenotypes? (c1) How does the environment interact with developmental processes?, (c2) How does environmental change influence phenotypic evolution?, and (c3) How does developmental evolution affect the environment?

We believe that artificial life (ALife) approach based on the evolution of virtual creatures can greatly contribute to eco-evo-devo and, therefore, started several studies taking the approach. Karl Sims is a pioneer in the use of artificial evolution of virtual creatures. In his model, a creature is composed of a hierarchy of 3D rigid parts. The morphologies of creatures and the neural systems for controlling their muscle forces are evolved simultaneously

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✉ Takaya Arita
arita@nagoya-u.jp

[1] Nagoya University, Nagoya, Japan

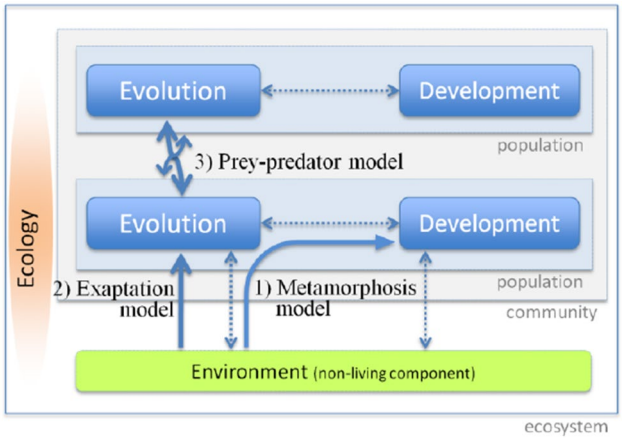

Fig. 1 Eco-evo-devo and the three models presented in this paper

using several fitness evaluation functions (e.g., swimming, walking and jumping) in a physically simulated 3D world [6]. He also used a competitive context in which pairs of creatures contend to gain control of a common resource. He successfully demonstrated the evolution of interesting and diverse strategies and counter-strategies [7]. Since the pioneering work of Sims, the evolution of morphology and behavior of virtual creatures have been studied further in of the field of artificial life. However, very few studies have focused on eco-evo-devo. This paper reports on the current status of our three lines of study focusing on eco-evo-devo, and thereby, presents the possibility of theoretical contribution of artificial life approach to eco-evo-devo.

Figure 1 illustrates those studies in the context of eco-evo-devo. The first model deals with the evolution of metamorphosis. We investigate whether metamorphosis emerges and, if so, what kind of changes occur during virtual organism's lifetime. To address some of the questions posed above including a3, b1 and c1, we have developed a model of fine-grained artificial development, in which organisms grow from a single cell, through subsequent divisions. The second study focuses on exaptation, which is defined as a shift during evolution in which a trait that had originally been shaped to serve a particular function or emerged as a byproduct of another adaptation later came to serve a different role [8]. We examine the process and the mechanism of exaptation triggered by an environmental change, using a simplification of Sims' model. In particular, we can investigate the above question c2 by performing evolutionary experiments with this model. Our third study addresses also question c2 but deals with prey-predator environments. Specifically, the focus of this study is the interaction between population and evolutionary dynamics.

In general, artificial development models can be classified into two approaches: grammatical and cell chemistry based [9]. The grammatical approach uses high level, algorithmic abstractions of development, and, have been shown to be successful in producing diverse and life-like morphologies of virtual creatures. The second and the third studies employ this approach. The cell chemistry based approach explicitly simulates interactions of a large number of independent cells, and have demonstrated the evolution of complex shapes or organized behavior but with higher computational costs. The first model employs this approach.

2 Metamorphosis model

2.1 Introduction

Metamorphosis is a process during which an organism that has already finished its embryonic development (e.g., a tadpole) undergoes a relatively fast and considerable change to its body structure (e.g., transforming into a frog), responding to an environmental change. Metamorphosis is exhibited by a wide variety of taxa as different as insects, mollusks or amphibians [10]. We consider metamorphosis as a successfully evolved developmental strategy in the sense that it allows evolution to produce a solution that can take two different forms, each adapted to its target environment.

The purpose of this study [11] is not only to offer new insights into eco-evo-devo but also to apply them to robotics [12]. There is an increasing interest in the use of soft and deformable structures in the robotic systems, as these are crucial for dealing with uncertain and dynamic environments. We believe that this study will contribute to the realization of the soft-robots that could efficiently operate in radically different environments and change their form on demand.

2.2 Model

2.2.1 Development and locomotion of creatures

Figure 2 shows a lifetime of a virtual creature. Developmental and evaluation phases are separated: the morphology of the creature emerges through a developmental process and is then tested for its gait performance as a simulated robot in a 2D physics based environment. Embryos grow from a single cell, through divisions and apoptosis, on a continuous 2D space. A copy of the same control network regulates each cell. Each cell also perceives spatial signals (maternal morphogens), and signals from their neighbors. The outputs of the network determine whether a cell divides and in what direction, or dies. Creatures are soft-bodied: made of elastic material and capable of actuation through contracting and expanding material locally, and are simulated using spring physics (point masses connected by

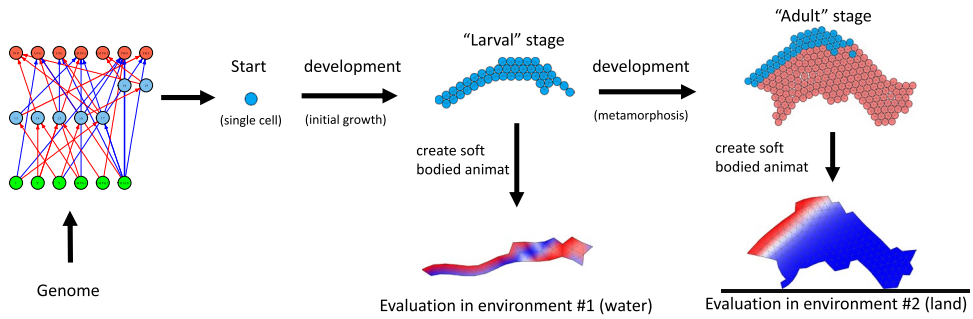

Fig. 2 Two stages each of which is composed of development and evaluation

springs + hydrostatic skeleton). A simple fluid drag model was used when the creatures are in the aquatic environment. Actuation occurs as each cell can increase or decrease the resting length of all of its springs, which is controlled by a sine wave oscillation of the resting length with a fixed phase, amplitude and frequency.

2.2.2 Fitness and evolution

We use the NEAT algorithm [13] to evolve networks that control development of each creature. We first allow each genome to control the embryonic development for some predefined number of time steps. The performance of an emerged morphology is then evaluated in the first (e.g. aquatic) environment (e.g. distance achieved during the swim). After that, development continues again for a set number of time steps. The performance is then evaluated in the second (e.g. terrestrial) environment (e.g. distance achieved by the walk). The fitness is a simple scalarization of distances achieved in its larval and adult stages.

2.3 Results

We evaluated developed individuals first in an aquatic environment, allowed them to continue growth and then evaluated them again in a terrestrial one. By analyzing repeated evolutionary runs, each initialized with a different random seed, we were able to observe that evolution converges on certain types of morphologies. It would typically produce robots with elongated, swimming larval stages that grow "legs" during metamorphosis so they can move on land by cyclically shifting their center of weight between them (Fig. 3).

We then performed the same experiments with the order of the environments inverted: larva had to move on land and adult had to move in the water environment. This time, we repeatedly observed how the evolution produces legged robots that shed their "tail" and remove their legs during metamorphosis so they can swim.

These results successfully show how it is possible to evolve soft-robots that are adapted to two different

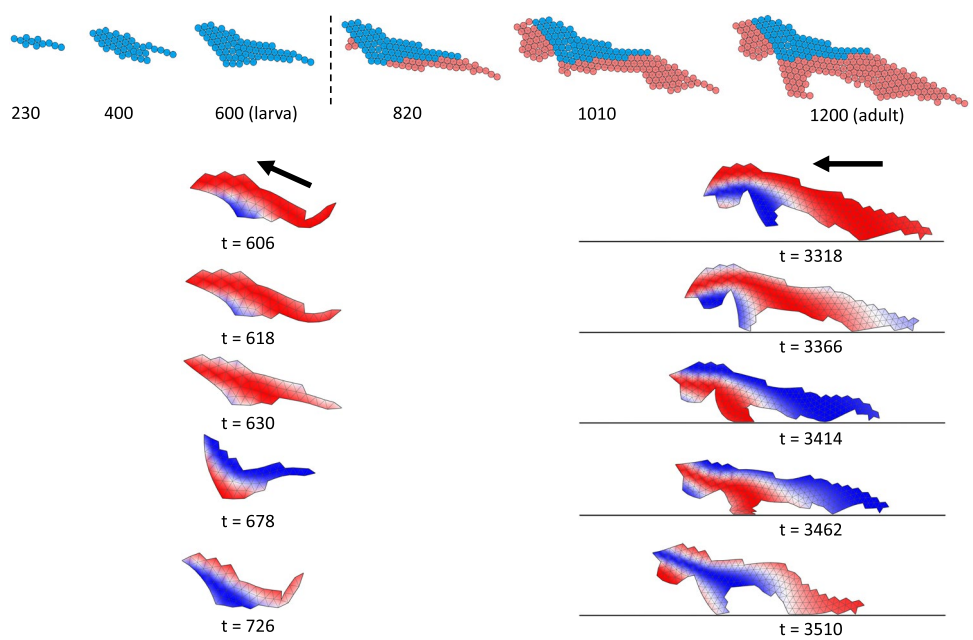

Fig. 3 An example of development (*top*) and swimming (*left*, *bottom*)/walking (*right*, *bottom*)

environments by transforming their morphology from one form to another, through a simple modification of a developmental system.

3 Exaptation model

3.1 Introduction

Exaptation describes a shift during evolution in which a trait that had originally been shaped for a particular function or as a byproduct of another adaptation later came to serve another role [8]. A famous example of exaptation is birds' feathers. Since early feathered dinosaurs were missing many other adaptations necessary for flight, we know that feathers initially served some other purposes, such as thermoregulation or signaling. It was only after that some groups of feathered dinosaurs exapted feathers for flight. In general, when organisms evolve and are shaped by natural selection, there will always be many forces that will prevent them from achieving an optimal design, e.g., historical contingencies, cost, or simply a lack of available variation. Exaptation can be considered as an important factor to deal with these constraints, and actually, it has been claimed that exaptation can be found at every level of biological organization including human culture [14].

The purpose of this study [15] is to understand the process and the mechanism of exaptation using artificial life approach. We use Morphid Academy [16], which is an open source simulation software to evolve virtual creatures in a 3D physically simulated environment. The virtual creature model is a simplification of Sims' model. The simplification in body and neural structure reduces the evolutionary search space, and has been demonstrated to perform well for various evolutionary tasks.

3.2 Model

3.2.1 Expression of creatures

Each virtual creature is composed of several 3D rectangular solid body parts connected with simple hinge joints. Neuron nodes are stored distributedly in the body parts and control the behavior of the creature by working as an artificial neural network. Each node corresponds to either a sensor measuring the relative angle and a distance of the creature belonging to the other species nearest to the creature, a McCulloch–Pitts neuron with hyperbolic tangent transfer function, a neuron generating a sinusoidal periodic signal, and an actuator neuron outputting a value determining the power to a hinge joint.

The genotype of creatures is described by a directed graph composed of body part nodes, each of which corresponds to a body part, and the links, each of which corresponds to a joint. To express the repetitive structure, it can have recursive links returning to themselves, and duplicative links sharing the same source and destination nodes. The initial node is called a root node, and the corresponding body part is called a main part. Each body part node has (a part of) a neural network.

The conversion from genotypic to phenotypic representations corresponds to a development process, during which recursive and duplicative nodes are expanded by generating nodes and links. The resultant graphs describe the body structure of the creature by interpreting the body part nodes as body parts, and links as hinges. During the development process, information of the neural network is also duplicated when nodes are duplicated. The links in the other nodes are also duplicated, which have a node in the original node as a destination node. Recurrent links are allowed in the neural network.

3.2.2 Evolution of creatures

A standard steady-state genetic algorithm is used with deterministic tournament selection. In each generation, three individuals are selected from the population and they are evaluated to obtain fitness function values. After that, with some probability, the worst virtual creature is just replaced by the best one. Otherwise, crossover or grafting is applied to the best and second-best creatures to produce a child, which will replace the worst creature. Mutation is then applied to the resulting child creature with a small probability. There are two types of mutation of genotype graphs: adding and removing genotype nodes and links, and adding variation to the parameters of them. Mutation to neural networks is performed also by adding and removing neurons and neural links, and adding variation to the parameters of them.

In the evaluation phase, the creatures act for a certain time in a three-dimensional physically simulated environment with gravity and friction. The creatures' actions are evaluated by a fitness function. We use three kinds of fitness functions: (1) moving distance on a flat terrain, (2) moving distance on a rugged terrain, and (3) jumping height on a flat terrain.

Each exaptation experiment is composed of two stages (Fig. 4). In the first stage, randomly generated creatures evolve for a fitness function. In the second stage, the initial population consists only of copies of the best evolved creature in the first stage and they evolve for another fitness function. This replacement of fitness function corresponds to an environmental change. In the second stage, evolutionary operators that affect the topology of genotype graph are

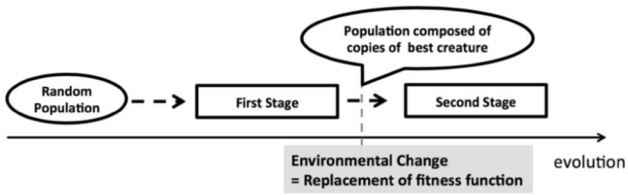

Fig. 4 The course of evolution composed of two stages

excluded to constrain the evolution. This evolutionary constraint is introduced to focus on exaptation in the second stage.

3.3 Results

In the first stage, virtual creatures evolved distinctive strategies successively for each fitness function. For fitness function 1, they evolved to move by pushing against the ground with a few large body parts. For fitness function 2, they evolved typically a long body composed of many parts, to walk on the rugged terrain without being trapped by the gaps. For fitness function 3, they evolved a snake-shape body form and used it like a spring to jump.

It should be noted that in the second stage of the exaptation experiments, the creatures seemed to expt their strategy that had been evolved for the fitness function in the first stage. For example, a typical creature evolved in the second stage for fitness function 2 moved on rugged terrain utilizing jumping behavior which had been evolved in the first stage for fitness function 3. In another example, the creatures with multiple legs evolved for fitness function 2 in the first stage exapted their legs for movement by hopping behavior for fitness function 1 in the second stage.

Overall, the average fitness of the best creatures evolved in the normal experiment is the highest for every fitness function. However, it is a remarkable fact that exaptation experiments for fitness function 2 evolved the best and the second-best creatures among the best creatures evolved in all of the experiment trials. It suggests that exaptation has a potential to deal with difficult tasks. We also proposed measurements for the change of morphology and behavior to examine the exaptation of traits evolved in the first stage quantitatively.

4 Prey-predator model

4.1 Introduction

Recent studies have reported that population dynamics and evolutionary dynamics, occurring at the different time scales, can be affected by each other (eco-genetic feedback [17], eco-evolutionary feedback [18]). The purpose of this study [19] is to explore the interaction between population and evolutionary dynamics. Here, we also employ the Morphid Academy environment.

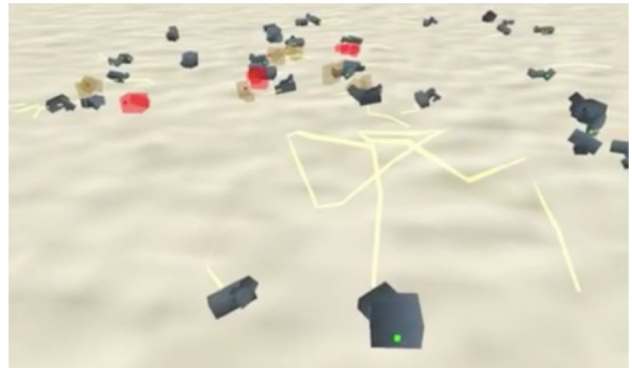

Fig. 5 Predators and prey in the simulated world

4.2 Model

To represent the interaction between the group of predators and prey, we simulate and evaluate all predator and prey creatures from both of the population pools in a shared environment (Fig. 5). The methods of genetic and phenotypic representations are the same as those in the exaptation model.

The prey that have successfully escaped predation until the end of the simulation obtain an evaluation value which depends on its volume, representing the cost for the maintenance of the body size, while the predators that captured more prey (and, in addition, moved closer to a prey when the simulation ended) can obtain a larger evaluation value. Two populations representing the predators and the prey are concurrently evolved using a genetic algorithm. Both population sizes are changed simultaneously, depending on the performance of the creatures.

4.3 Results

Here we show the results of the experiments in which the predators were not allowed to evolve so as to understand the basic dynamics. We pre-evolved predators in the preliminary experiments with random prey, and some successfully evolved predators were used to seed the initial predator population.

We found the different interactions between population and evolutionary dynamics at the short and long timescales. When we focused on the short-term dynamics, we observed a simple cyclical dynamics of the population of predators and prey, which corresponds to a Lotka-Volterra type population dynamics (Fig. 6). This is because, the population dynamics was too fast for the evolutionary dynamics

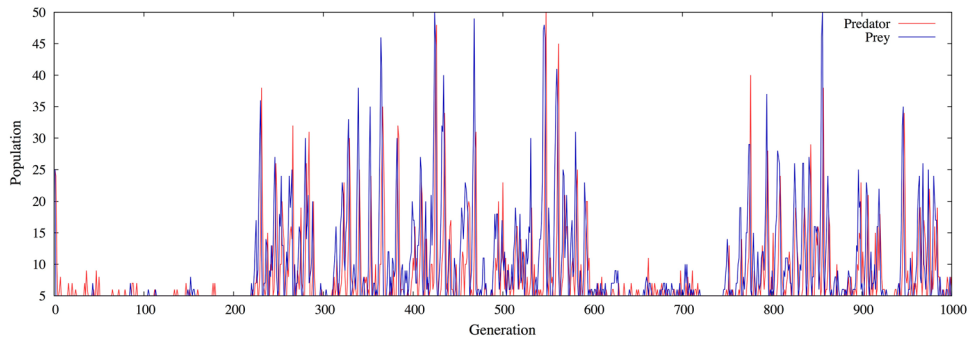

Fig. 6 A typical transition of population size (*red* predators, *blue* prey)

to adapt to. In contrast, when we focused on the long-term dynamics, we observed the complex interactions between the population dynamics of both species and the evolutionary dynamics of the trait of prey. Specifically, we found the inverse correlation between the population sizes and the average volume of the prey, and their continual fluctuations, yielding the emergence of defensive and non-defensive morphological strategies of prey (Figs. 7, 8). This is due to the fact that the target of selection for the prey switched between defense against predation and reduction in the cost of defense depending on the population size of predators.

In general, understanding the mechanism of extinction based on the predator–prey relationship is important to the conservation and understanding of ecosystems. As a next step of this study, we focused on the sudden extinction in this model. We found that an increase in the volume increases the cost but promotes good capturing/defensive strategies [20]. The predator which has good strategy and large body can pay its maintenance cost through successful predation when the population density is high. However, it will fail to do so when the population density is low. Therefore, we estimated that the arms race of the predation/escape

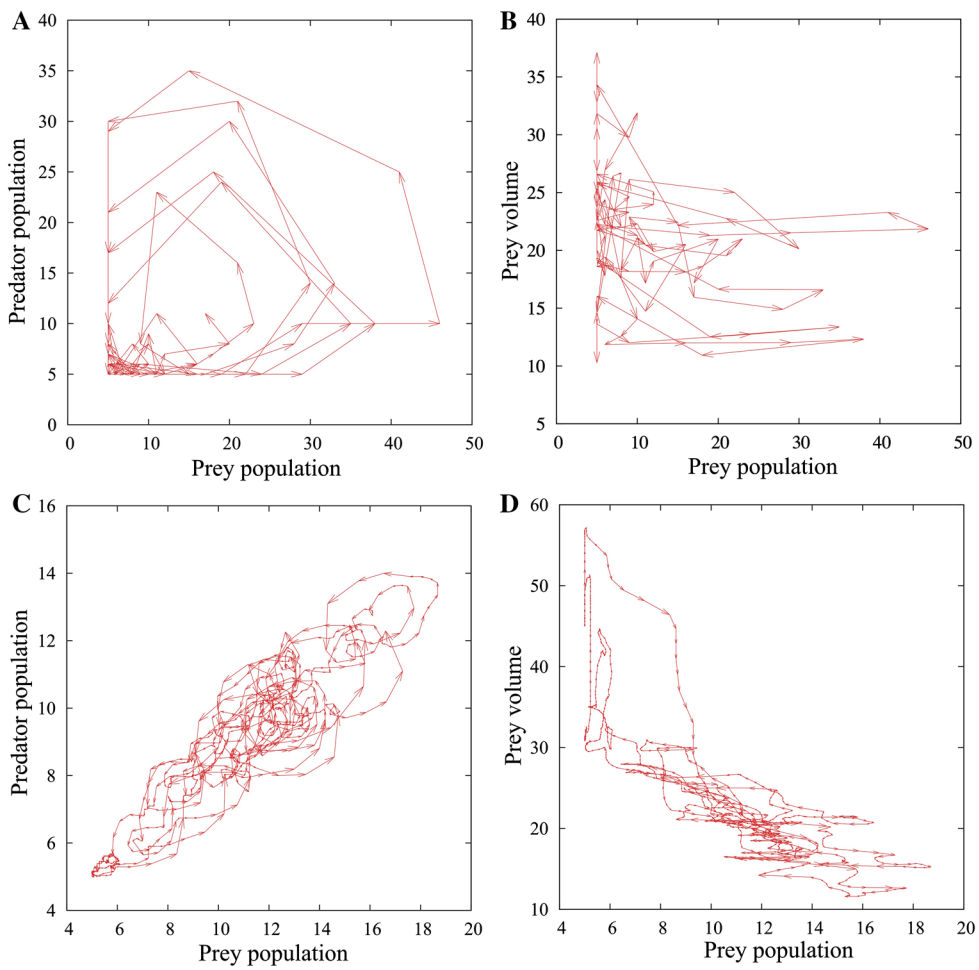

Fig. 7 Typical trajectories of community and evolutionary dynamics. **a** The predators and prey populations in short-term dynamics. **b** The volume and population of the prey in short-term dynamics. **c** The predators and prey populations in long-term dynamics. **d** The volume and population of the prey in long-term dynamics

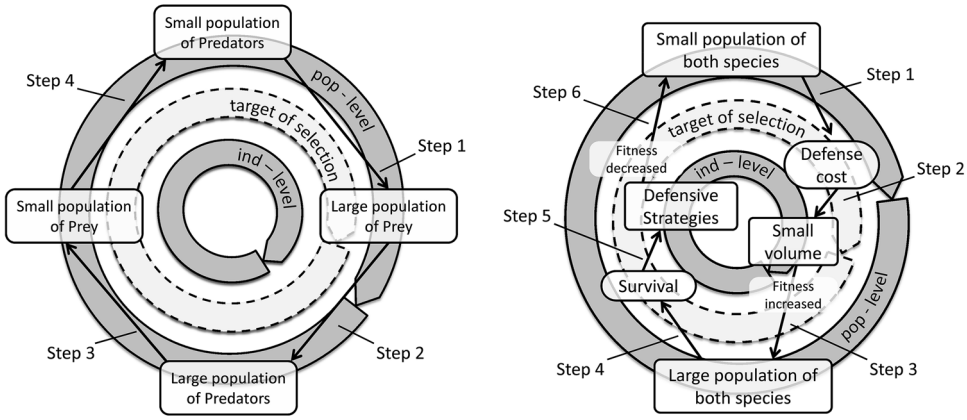

Fig. 8 The interactions between the community and evolutionary changes in the short-term (*left*) and long-term (*right*) dynamics. The short-term dynamics represents a typical prey-predator population dynamics, like Lotka-Volterra model, independent of the trait evolution. On the other hand, the long-term dynamics represents the interaction between population and evolutionary dynamics as follows. When the population size of each population is small, predation events are rare. In this situation, the defense cost becomes the target of the selection. Therefore, creatures with small volume increase in the population. This means that the evaluation value, in other words, the number of children increases. The increase in population size of each species means the increase of predation events. It makes defensive strategies become the target of the selection. Because of the high cost for their large volume as well as the high predation pressure, the fitness of the prey then decreases. Thus, the population of the prey decreases, which further decreases the population of predators. This process generated by the coupling of population dynamics and evolutionary dynamics is repeated

ability causes an increase in extinction possibilities when the population density is low in Lotka-Volterra system. This prediction method based on the calculation of the frequency of the evolutionary changes might be used not only in artificial life frameworks but also in natural ecosystems.

5 Conclusions

Ecological evolutionary developmental biology, called eco-evo-devo, attempts to reveal the mechanisms associated with the interactions among development, evolution and environment. This paper reports on our ongoing artificial life studies on eco-evo-devo based on the evolution of morphology and behavior of virtual creatures. Artificial life approach based on the evolution of virtual creatures can greatly contribute to eco-evo-devo by offering the synthetic and its computational methodology, and explicit attention to emergent phenomena.

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